PEANO REMAINDER OF TAYLOR EXPANSION

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Theorem 1 (Taylor expansion with Peano remainder)**.** *Assume f is defined in a neighborhood of $\mathbf{0} \in \mathbb{R}^n$ and f is secondly differentiable at $\mathbf{0}$, i.e. $\frac{\partial f}{\partial x_i}$ is differentiable at $\mathbf{0}$ for $i = 1, 2, \cdots, n$.*

$$f(\boldsymbol{x}) = f(\mathbf{0}) + \nabla f(\mathbf{0}) \cdot \boldsymbol{x} + \frac{1}{2}\sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}(\mathbf{0}) x_i^2 + \sum_{1 \le i < j \le n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) x_i x_j + R(\boldsymbol{x})$$

with $R(\boldsymbol{x}) = o(|\boldsymbol{x}|^2)$ as $\boldsymbol{x} \to \mathbf{0}$.

Proof. By definition, for $i = 1, 2, \cdots, n$

$$\frac{\partial f}{\partial x_i}(\boldsymbol{x}) = \frac{\partial f}{\partial x_i}(\mathbf{0}) + \nabla\left(\frac{\partial f}{\partial x_i}\right)(\mathbf{0}) \cdot \boldsymbol{x} + R_i(\boldsymbol{x}) \tag{1}$$

with

$$\lim_{\boldsymbol{x} \to \mathbf{0}} \frac{R_i(\boldsymbol{x})}{|\boldsymbol{x}|} = 0, \quad i = 1, 2, \cdots, n \tag{2}$$

Then,

$$\begin{aligned}
f(\boldsymbol{x}) &= f(\mathbf{0}) + [f(x_1, 0, \cdots, 0) - f(0, 0, \cdots, 0)] + [f(x_1, x_2, \cdots, 0) - f(x_1, 0, \cdots, 0)] \\
&\quad + \cdots + [f(x_1, x_2, \cdots, x_n) - f(x_1, \cdots, x_{n-1}, 0)] \\
&= f(\mathbf{0}) + \int_0^{x_1} \frac{\partial f}{\partial x_1}(t, 0, \cdots, 0)\mathrm{d}t + \int_0^{x_2} \frac{\partial f}{\partial x_2}(x_1, t, \cdots, 0)\mathrm{d}t + \cdots + \int_0^{x_n} \frac{\partial f}{\partial x_n}(x_1, \cdots, x_{n-1}, t)\mathrm{d}t \\
&= f(\mathbf{0}) + \int_0^{x_1} \left(\frac{\partial f}{\partial x_1}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1^2}(\mathbf{0})t + R_1(t, 0, \cdots, 0)\right)\mathrm{d}t \\
&\quad + \int_0^{x_2} \left(\frac{\partial f}{\partial x_2}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{0})x_1 + \frac{\partial^2 f}{\partial x_2^2}(\mathbf{0})t + R_2(x_1, t, \cdots, 0)\right)\mathrm{d}t \\
&\quad + \cdots \\
&\quad + \int_0^{x_n} \left(\frac{\partial f}{\partial x_n}(\mathbf{0}) + \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{0})x_1 + \cdots + \frac{\partial^2}{\partial x_{n-1}\partial x_n}(\mathbf{0})x_{n-1} + \frac{\partial^2 f}{\partial x_n^2}(\mathbf{0})t + R_n(x_1, \cdots, x_{n-1}, t)\right)\mathrm{d}t \\
&= f(\mathbf{0}) + \nabla f(\mathbf{0}) \cdot \boldsymbol{x} + \frac{1}{2}\sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}(\mathbf{0}) x_i^2 + \sum_{1 \le i < j \le n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{0}) x_i x_j + R(\boldsymbol{x})
\end{aligned}$$

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Here the remainder

$$\begin{aligned} R(\boldsymbol{x}) &= \int_0^{x_1} R_1(t, 0, \cdots, 0)\mathrm{d}t + \int_0^{x_2} R_2(x_1, t, \cdots, 0)\mathrm{d}t + \cdots + \int_0^{x_n} R_n(x_1, \cdots, x_{n-1}, t)\mathrm{d}t \\ &= \int_0^{x_1} \frac{R_1(t, 0, \cdots, 0)}{t} t\mathrm{d}t + \int_0^{x_2} \frac{R_2(x_1, t, \cdots, 0)}{\sqrt{x_1^2 + t^2}} \sqrt{x_1^2 + t^2}\mathrm{d}t + \cdots \\ &+ \int_0^{x_n} \frac{R_n(x_1, \cdots, x_{n-1}, t)}{\sqrt{x_1^2 + \cdots + x_{n-1}^2 + t^2}} \sqrt{x_1^2 + \cdots + x_{n-1}^2 + t^2}\mathrm{d}t \end{aligned}$$

By (2), we know $\forall \epsilon > 0$, there exists $\delta > 0$ such that for all $\boldsymbol{x} \in [-\delta, \delta]^n$,

$$\left| \frac{R_i(\boldsymbol{x})}{|\boldsymbol{x}|} \right| < \epsilon, \quad i = 1, 2, \cdots, n$$

thus

$$\begin{aligned} |R(\boldsymbol{x})| &\le \epsilon \left(\left| \int_0^{x_1} t\mathrm{d}t \right| + \left| \int_0^{x_2} \sqrt{x_1^2 + t^2}\mathrm{d}t \right| + \cdots + \left| \int_0^{x_n} \sqrt{x_1^2 + \cdots + x_{n-1}^2 + t^2}\mathrm{d}t \right| \right) \\ &\le \epsilon \left(\frac{1}{2}x_1^2 + |x_1 x_2| + \frac{1}{2}x_2^2 + \cdots + |x_n|(|x_1| + \cdots + |x_{n-1}|) + \frac{1}{2}x_n^2 \right) \\ &\le n\epsilon|\boldsymbol{x}|^2 \end{aligned}$$

Therefore, $R(\boldsymbol{x}) = o(|\boldsymbol{x}|^2)$. □

Similarly, by Mathematical Induction, we could derive the Peano remainder for higher-order Taylor expansion.

Corollary 2. *Assume f is secondly differentiable at $\boldsymbol{x}_0$, then*

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{x}_0) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\boldsymbol{x}_0)$$